A literature review of aerospace engineering

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This paper examines the aerospace engineering field and sub-fields within aerospace engineering. We explore five sub-fields related to the subject. Aeroacoustics, investigating the generation and control of noise in aerodynamic systems; Aeroelasticity, exploring the interactions between structural dynamics and aerodynamics; Combustion and Propulsion, driving the advancement of air and spacecraft propulsion systems; Systems design, integrating the components into a reliable and efficient system; and Uncrewed Aerial Vehicles (UAVs), revolutionizing the aerospace field with autonomous and remotely piloted vehicles. By examining the sub-fields, we aim to comprehensively understand of the history, current state, and future of aerospace engineering.

I. INTRODUCTION

Aerospace engineering, a field synonymous with cutting-edge technology and innovation, is the field of engineering interested in the design, development, construction, and operation of vehicles operating in Earth's atmosphere or in outer space [1]. We have already embarked on an era embodied by continuous advancements in aviation and space exploration. As such, it becomes imperative to review the existing literature within aerospace engineering. This review will center around the vehicles operating in the Earth's atmosphere and encompass a variety of sub-fields within aerospace engineering: aeroacoustics, aeroelasticity, combustion and propulsion, systems design, and uncrewed aerial vehicles (UAVs). By examining the wealth of knowledge available in these sub-fields, this review seeks to understand the history and current state of aerospace engineering as well as where we could be headed in the future.

II. HISTORY

The earliest ideas of human flight can be traced back to Greek mythology and the story of Icarus and his father, Daedalus [2]. They attempted to escape from the labyrinth of Crete by creating and attaching wings to their arms in an attempt to mimic those of birds. Constructed of feathers, wax, and threads, they attempted their escape leading to the story of Icarus flying too close to the sun; the wax melted his wings, and he ultimately drowned as sea [3]. In terms of mechanical flying machines, flight can be traced back to a Renaissance visionary, Leonardo da Vinci. His notebooks date back to the 15th century [2]. Among da Vinci's sketches are flying machines and aeronautical contraptions. For example, an ornithopter with mechanically driven wings to mimic the flight of birds [2] which can be seen in fig. 1.

FIG. 1. This image shows a modeled image of what a sketch of an ornithopter looked like. Image is courtesy of the National Air and Space Museum.

Progressing to France in 1783, we find the first vehicle capable of sustaining flight while carrying a human, the hot-air balloon [4]. The early balloon flights were witnessed by one, Benjamin Franklin. Franklin predicted that balloons would soon be used for tactical purposes such as surveillance [2]. He later proposed concepts for lighter-than-air concepts combined with propulsion which led to the development of lighter-than-air aircraft with rigid internal frames that became known as *dirigibles*[2]. However, the invention of lighter-than-air vehicles occurred independently of the development of the aircraft [4]. The breakthrough in aircraft development occurred when Sir George Cayley drew an airplane incorporating a fixed wing, an empennage (consisting of horizontal and vertical tail surfaces for stability and control), and a separate propulsion system [4]. The first *dirigible* was built and flown in 1852 by Henri Giffard [2].

Half a century later the Wright brothers became the first to fly a powered biplane that took off and landed under the control of its pilot. After this, advances in aircraft development and aerospace engineering skyrocketed. Less than a decade and a half later, during WWI, biplanes were being used for military missions like bombing, dog fights, and reconnaissance. Following World War I, advancements focusing on civil air transportation were made. This included things such as Charles Lindbergh's first solo nonstop transatlantic flight and the twin-engine monoplane design of the Boeing 247-D [5] entering service [4]. During and after World War II, many strides were made in aeronautical engineering, leading to the development of increasingly larger and faster aircrafts that benefited both military and civil aviation [2]. The late 1950s and '60s saw a period of intense growth for astronautical engineering, as the U.S.S.R. orbited Sputnik I, the first artificial satellite in 1957. The following year, the *Boeing 707* airliner was introduced to airline service and could make transatlantic and transcontinental flights [5]. From there the "space race" commenced, leading to the US being the first to land a man on the moon in 1969. By 1975, American manned missions to the moon had ceased. Since then, we began to explore further out in the universe with unmanned missions to many of our solar system's planets.

III. AEROACOUSTICS

Aeroacoustics is the branch of science that encompasses the study of sound generated by aerodynamic flows [6]. This branch rose in importance when jet-powered flight became increasingly more common as it involves reducing harmful vibrations, structural fatigue, and noise pollution. Initial concepts and early models of jet-powered civilian airplanes caused more research to be put into this specific branch of aerospace engineering. The first major challenge the field faced was in the development of the *Concorde* in the early 1970s [7]. This plane called for noise reduction for the supersonic aircraft while minimizing the loss in thrust performance [7]. This first era of intense research regarding aeroacoustics can be called the golden age of aeroacoustics [7]. Since then the field of aeroacoustics has only grown in size and complexity. As aircrafts evolved, aeroacoustics did the same. As air travel became more accessible, people began to complain about noise more frequently. That called for more research into the components of the air frame and how they contribute to noise while landing [7]. Recently, more people have pushed for things like the silent aircraft initiative [8], the strive towards more environmentally responsible aircraft which in turn cause more complex design problems when it comes to noise reduction while improving other aspects of the aircraft.

The physics behind aeroacoustics is rather complex as it deals with fluid dynamics. The concepts in fluid dynamics have multiple graduate level courses on how to deal with and solve equations related to them as well as textbooks hundreds of pages long. For the sake of this paper we will only scratch the surface of the complexity as depth that aeroacoustics involves. Slowly changing turbulence is rather inefficient, however sharp geometrical changes can result in a transfer of disturbance energy from otherwise non-radiating convected modes to those than radiate [9]. We can see an example noise sources in a turbofan engine in fig. 2.

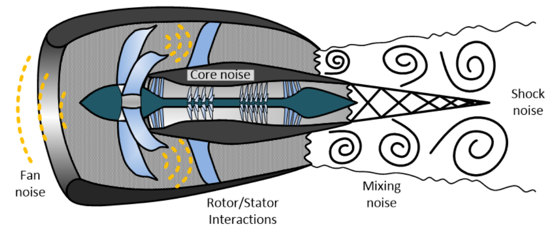

FIG. 2. This image shows what kind of sounds are generated in a typical turbofan engine as well as where the sounds are generated. Image courtesy of the University of Florida.

For nonlinear sound generation, techniques such as Direct Numerical Simulation (DNS) [10] and Large Eddy Simulation (LES) [11] of turbulent flow have significant overlap since the flow is governed by the unsteady compressible Navier-Stokes equations [12]. When it comes to attempting to solve equations we cannot do the proper justice to the work many people have made their life's work. The best we can do in this paper is to recommend another article written by Hirschberg and Rienstra in 2004 [13].

We instead focus on the role that aeroacoustics play in aerospace engineering. Aeroacoustics play a direct role in aerospace engineering as it has a direct impact on design, performance and safety of the aircraft or spacecraft. Being able to understand and control the noise produced by the vehicles play a role in numerous ideas. The first idea being compliance with regulations. Loud aircraft can pose serious environmental concerns if the engines are too loud, especially while taking off and landing. Aeroacoustics aid researchers and designers in complying with the regulations which allows for less impact on the environment. This also directly ties into human comfort. The louder aircrafts are the more of a problem the aircrafts become. For both passengers riding in the aircrafts as well as the communities that live around airports. Aeroacoustics help limit noise production to keep passengers and the communities as happy as they could be. On the opposite side of civilian life, aeroacoustics are also vital to military operations. The study of aeroacoustics plays a giant role for the military. The prominent example for the military are the stealth operations. Without aeroacoustics to reduce the noise production and scatter the noise that is produced stealth missions would be very difficult if not impossible.

IV. AEROELASTCITY

Aeroelasticity is a relatively vague term because it encompasses many different phenomena that include elastic forces as well as inertial and aerodynamic forces [14]. For the sake of this paper we will define aeroelasticity as phenomena involving interactions among elastic, aerodynamic, and inertial forces. Issues in aeroelasticity did not arise to a noteworthy role until the early stages of World War II and the use of jet-powered aircrafts [15]. Prior to this time, aircrafts were powered by propellers and had relatively low top speeds. That meant load requirements on the structures of the aircrafts allowed designs to be sufficiently rigid [15] which can be seen by the use of the biplanes during World War I. Aeroelasticity problems arose when designers moved away from the biplane in favor of the monoplane designs [15]. An early example of this kind of problem can be seen in the development of the Fokker D-8 airplane. They had been put into production after the torsional stiffness was determined by standards for biplanes. Within days of their initial combat action, wing failures occurred repeatedly at high-speed dives [15]. Anthony Fokker, the designer, discovered that load from air pressure increased faster at the wing tips than in the middle during a steep dive [15]. This discovery can be largely seen as the commencement of serious aeroelasticity research.

For the sake of this paper, we will consider only one facet of aeroelasticity, flutter. Flutter can be caused as the airplane deforms under a load. The interacting feedback between deformation and load on the aircraft may lead to an often destructive self-excited oscillation wherein energy is absorbed from the airstream, flutter [16]. The establishment of flutter depends directly on stiffness, and only indirectly on the strength of the airplane, analogous to the slope of the lift curve rather than on the maximum lift [16]. To simplify flutter further for the constraints of this paper we will examine subsonic flutter. We are able to produce a full continuum aeroelastic model from models developed by Goland, Ashley, and Tzou [17–19]. First it will be useful to have a model of a wing which can be seen below.

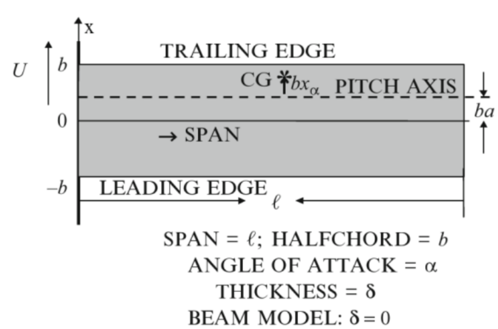

FIG. 3. Model of a wing courtesy of source [20].

The bending dynamics of the wing as described by an Euler equation can be written as

$$\begin{aligned} m\ddot{h}(t,y) + S\ddot{\theta}(t,y) + EIh^{''''} &= L(t,y), \\ t > 0; 0 < y < 1 < \infty, \end{aligned} \tag{1}$$

and the torsion by the Timoshenko equation:

$$I_\theta \ddot{\theta}(t,y) + S\ddot{h}(t,y) - GJ\theta^{''}(t,y) = M(t,y), \tag{2}$$

where the forcing terms: $L(t,y)$ is the aerodynamic force per unit length (lift); $M(t,y)$ is the applied aerodynamic moment about the elastic axis per unit length[20]. The moment about the elastic axis per unit length are both determined later from the airflow model.

- m = Mass per unit length
- I_θ = Cross-sectional moment of inertia
- EI = Bending Stiffness
- GJ = Torsional Stiffness
- S = Coupling constant $= mbx_\alpha, |x_\alpha| < 1$,
- see fig. 3 for x_α; all constant, justifying the term “uniform.” [20]

Here x_α can be positive or negative, and it is required that $S^2 < mI_\theta$ [20]. The convention throughout is that the superdots denote the partial derivatives with respect to time and the superprimes are the partial derivatives with respect to the spatial y coordinate. Through a very lengthy process as well as many other equations that can be found in [20], we can analyze flutter in the wings. Unfortunately, every article we could find was behind a paywall to access the necessary parts of the article to dive deeper into the mathematical equations related to flutter and the analysis to minimize said flutter.

Instead we jump straight into the relationship that flutter, more broadly aeroelasticity, has on the aerospace engineering field. As we have established aerospace engineering encompasses the design and improvements for both aircrafts and space crafts. Aeroelasticity is more of a focus for aircrafts rather than spacecrafts due to the nature of air pressure that aircrafts always operate under. Although spacecraft still have to follow some aeroelasticity rules as they launch out of our atmosphere. The role aeroelasticity plays in aerospace engineer is nothing short of crucial. Without it, aircrafts and spacecrafts alike would crumple under torsional stress. If we did not pay attention to aeroelasticity and its effects then there would be a high likelihood that spacecrafts and jet-powered aircrafts would have never been possible due to the high rates of speed that would have cetrainly caused damages and possibly death.

V. COMBUSTION AND PROPULSION

Combustion can be considered the oldest technology of mankind and arguably our most important discovery/invention. Combustion refers to the process of burning, where a fuel reacts with an oxidizer to produce heat and usually light. Meanwhile propulsion involves the generation of thrust to propel an object through a fluid (typically air). For the sake of our paper, combustion will be the propulsion we refer to. Combustion as a form of power can be traced back to the early 18th century to a man named Thomas Newcomen. He built a piston-and-cylinder steam-powered water pump for use in mines which can be seen in fig. 4. Newcomen created the first practical device to harness the power of steam to produce mechanical work [21].

Then, at the end of the 18th century a man named Robert Street conceptualized the first internal combustion engine [21]. Around 40 years later, in 1837, the first American patent for an electric motor was patented [21]. Then came electric power plants, gasoline-powered cars, the first flight by the Wright brothers, and the maiden flight of the Hindenburg all within the next hundred years following. We can see that throughout history there have been major technological advancements related to combustion. More recently, the use of combustion as a form of propulsion has become increasingly more popular.

In terms of the physics associated with combustion we have to follow all the conservation laws. First we have to obey the conservation of mass. Applying the conservation of mass to a fluid passing through an infinitesimal, fixed-control volume, the following equation is obtained:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i) = 0, \tag{3}$$

where ρ is fluid density, u is velocity, and t is time [22]. The next conservation that combustion follows is the conservation of momentum. Where for a fluid passing through an infinitesimal, fixed-control volume, the following equation is obtained:

$$\frac{\partial}{\partial t}(\rho u_j) + \frac{\partial}{\partial c x_i}(\rho u_i u_j) = \rho g_j + \frac{\partial \delta_{ij}}{\partial x_i}, \tag{4}$$

where the stress tensor δ_{ij} represents the external stresses, consisting of normal and shearing stresses [22]. The final conservation law combustion must abide by is the conservation of energy. The First Law of Thermodynamics states that the increase in energy in the system is equal to the heat added plus the work done on the system [23]. If we assume the system is incompressible and has a constant coefficient of thermal conductivity, then the energy equation is given by:

$$\rho \frac{De}{Dt} = \lambda \nabla^2 T + \Phi + S_h, \tag{5}$$

where

$$\begin{aligned}\Phi = \eta \Bigg[& 2\left(\frac{\partial u}{\partial x}\right)^2 + 2\left(\frac{\partial v}{\partial y}\right)^2 + 2\left(\frac{\partial w}{\partial x}\right)^2 \\ & + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}\right)^2 \\ & + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right)^2 - \frac{2}{3}\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right)^2 \Bigg].\end{aligned} \tag{6}$$

The last term, S_h, includes the heat of the chemical reaction, radiation, and any inter-phase exchange of heat [23]. This analysis of combustion reactions continues for other, less simplified, scenarios which can be explore more in depth in another article here: [23].

When it comes to relevance in regards to aerospace engineering combustion and propulsion are the bread and butter of any project for aircraft or spacecraft. Any movement aerospace engineering wants the craft to have needs to have a way to move. Propulsion is the answer. Now, combustion may not always be the way they achieve propulsion. However, it is arguably the most utilized form of propulsion as every diesel and gasoline-powered car has an internal combustion engine that propel the car forward. Now, aerospace engineers need a way to tie all of these complex systems that go into creating an air or space craft.

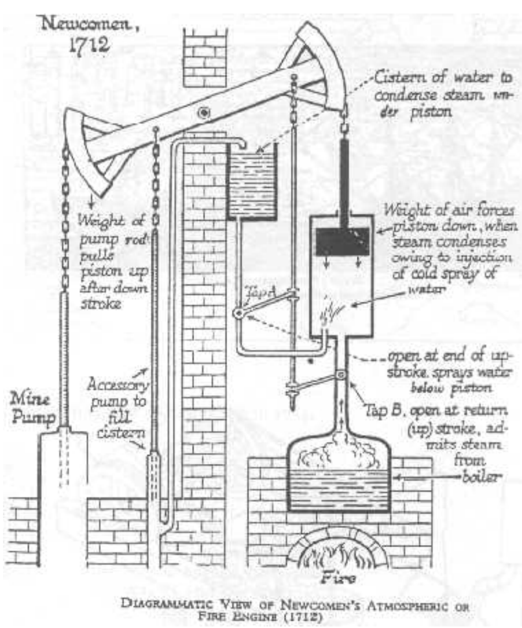

FIG. 4. This image shows the Newcomen's steam engine which was driven by the vacuum created when steam was condensed back into water. Image is courtesy of Weber State University.

VI. AEROSPACE SYSTEMS DESIGN

Aerospace systems in the modern world are becoming increasingly more complex and interconnected, although

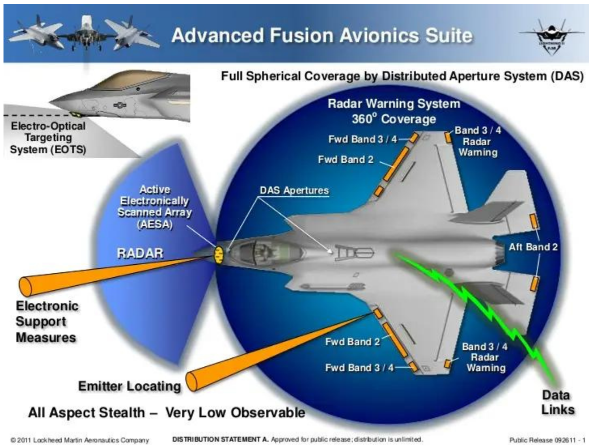

FIG. 5. This image shows some of the avionics suite that the *F-35* has. The image is courtesy of Lockheed Martin.

they are still required to meet rigorous requirements in performance, reliability, and safety requirements. As advancements are made, this field is incorporating more collaboration from multiple disciplines and their research. All of which are implementing rigorous systems engineering principles and analytical methods to meet the strict criteria. The aerospace system for any given project is how all the moving parts are connected. In today's day and age that is all done electronically through user interfaces (UIs) that have the ability to control almost anything in the system. A great example of the complexity of a modern system is in that of the *F-35*, a fifth generation tactical fighter. The fighter has an estimated 100 times more parameters to identify a threat than the previous generation [24]. A brief synopsis of some of the avionics can be seen in fig. 5.

The advancements in the *F-35* are thanks to more components and greater coupling between them. The *F-35* has 130 subsystems, 10^5 interfaces, and 90% of its functions are managed by software [25]. Compared to the fourth generation fighter that has 15 subsystems, 10^3 interfaces, and less than 40% of its functions managed by software [25].

The interesting nature of the complexity of systems led scientists to find ways to measure the complexity of a system. We see some of these measures which are given by [25]:

- **Level of Abstraction:** This denotes the visualization of the system at different levels of detail.

- **Type of Representation:** At each level of abstraction the system can have different representations. For example, a system can represented by a structural graph [26].

- **Size:** This is representative of the number of components and interactions within the system. Generally, the greater the number of components the higher the complexity in a system.

- **Heterogeneity:** The greater the heterogeneity of the components and interactions more is the complexity of the system.

- **Coupling:** There are two types of coupling in a system, direct and indirect. Direct coupling is the result of interdependency between the components due to direct physical connection between them. Indirect coupling occurs when a path with one or more components connects two components. This allows for a component to affect another without a direct physical connection.

Extending the concept of coupling, we can calculate coupling complexity using the following equation [25].

$$C_{cc} = \sum_{s=1}^{c} \left(n_s \sum_{i=1}^{n_s} W_{is} \right) + \sum_{k=1}^{m} W_k, \qquad (7)$$

where c denotes the number of feedback loops; n_s denotes the number of links in the sth feedback loop; W_{is} denotes the weight of the ith link of the sth feedback loop; m is the number of links that are not the part of any feedback loop. W_k denotes the weight of the kth link that is not the part of any feedback loop [25]. The benefit of this equation is that it automatically incorporates the size complexity, which increases with the number of components and interactions. By accounting for direct coupling, we ensure that the coupling complexity increases with the number of links [25]. This form of complexity measures extend to subsystems as well. This is explore more in depth in Tamaskar [25].

As aerospace engineering evolves, the system designs have to evolve to fit the criteria of performance, reliability, and safety. The advances in the system designs will only cause systems to become increasingly more complex. The systems in aerospace projects are what tie the entire project together. Without the ability to control multiple aspects of the projects from one system is vital. The system is the culmination of every department and researcher's work for any given project. The future of systems designs lie in the capability of the systems that run any given aerospace project. This can be seen in the next section on UAVs. The complexity and size of the systems that are required to run a system such as this will only continue grow as we progress to the future.

VII. UNCREWED AERIAL VEHICLES

Unmanned aviation developed in parallel with manned aviation. Several early pioneers of flight contributed to the development of both, including Glen Curtiss and Orville Wright [27]. Even though both styles of aviation were being developed simultaneously, the proportions of which they developed were not equal by any means. That was mainly due to insufficient technology. Development of an unmanned aircraft hinged on the conjunction of

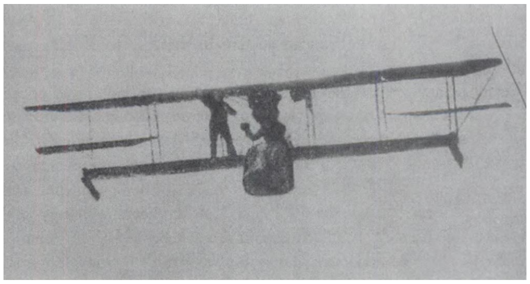

FIG. 6. Demonstration of Sperry Gyrostabilizer, Bezons, France, June 1914. Image is courtesy of the United States Air Force Museum.

three vital technologies in addition to powered flight: automatic stabilization, remote control, and autonomous navigation [27]. The first man to attempt to address all three in a single unmanned aircraft design was Elmer Ambrose Sperry [28]. Sperry's work with gyroscopes did not initially pique his interest in aviation. That is until he attempted to develop a gyrostabilizer for airplanes in 1909 [27]. His first model, a 30 pound gyrostabilizer was much too heavy as well as performed poorly when it encountered the three dimensions of flight [27]. However, when he reattacked the problem in 1911 he was able to create three smaller gyroscopes for the airplane's pitch, roll, and yaw [27]. He demonstrated his design outside of Paris in 1918 at France's Airplane Safety Competition [27]. An image of the demonstration of his stabilizer can be seen in fig. 6.

The demonstration, which was conducted by Sperry's son, won him first prize, 15,000 francs and the Collier Trophy for the most noteworthy achievement in aviation of 1914 [27]. In September of 1916, Sperry's system was tested before the US military. The demonstration consisted of the system being installed in a manned aircraft, whose pilot took off then engaged the autopilot, which then leveled off at its preset altitude, flew the preplanned course, then dove at the proper distance [27]. Then in 1939, a man named Reginald Denny introduced a low-cost remote controlled aircraft for training anti aircraft gunners, and in the same year he demonstrated another prototype for the U.S. Army [29]. Fast forward to nearly the end of World War II. In 1944, the first combat mission and use of a UAV was made when the U.S. Navy bombed Japanese positions using ten bombs [29]. Then around a decade later, in 1955, the first UAV was used for reconnaissance by the US military followed shortly by the British company Beechcraft [29]. Reconnaissance drones became essential during the Cold War after the Soviet Union shot down a *U-2* proving that high altitude is not enough protection for reconnaissance [30]. During the late 1970s and early ‘80s there was a huge push to aerospace research due to rapid electronic development [30]. During this time period the Arab-Israeli War was taking place. Israel invested a lot into its air force, ultimately leading to Israel being the leading country in UAV manufacturing [30]. The Israeli advancements during the electronic development paved a path for the U.S. to take the lead as the leading manufacturer of UAVs [30]. In the 21st century a basic requirement for UAVs has become stealth as well as integrating the on board system into one complex unit [30]. UAVs have now been designed as hunter-killers. That means the reconnaissance UAVs have the ability to attack if necessary and is how drone strikes occur. One can be seen in fig. 7.

The ties that the UAV has to aerospace engineering is monumental. The UAV represents the past, the present, and the future of aerospace engineering. Developing simultaneously as manned aircrafts, the concepts are intertwined throughout man's ability for powered flight. The current state of the world calls for so many reconnaissance drones because information is the most powerful thing someone can have on their enemy. The military is trying to limit the number of casualties from surveillance missions through the use of more UAVs, which has been done with some relative success. The future of aerospace engineering also includes UAVs. Especially within the Earth's atmosphere. The future of air superiority lies in UAVs as development and testing has begun on an unmanned fighter jet. A profession that during World War I killed over 52 thousand aircrew during combat [31] can potentially be reduced to zero. As for the future of civilian use. UAVs have as many applications as autonomous cars do. The possibilities are endless and have already begun to be explored, such as using UAVs to deliver packages like Amazon has experimented with. Another example being the little food delivery robots you see in large cities could be converted into UAVs. The possibilities really are endless for the future of UAVs and ultimately the future of aerospace engineering.

VIII. CONCLUSION

The literature review has provided an overview on some of key areas within aerospace engineering, shedding light on the variety of intricate details of this multidisciplinary field. The section on aeroacoustics dove into the role the noise generation and ultimately reduction strate-

FIG. 7. Image of a hunter-killer UAV. It is designed for stealth and reconnaissance, but also has missiles which can be seen on the underside of the UAV. Image is courtesy of the Wall Street Journal.

gies, acknowledging their significance to environmental concerns as well as passenger comfort and stealth capabilities. The exploration of aeroelasticity emphasized the history and the importance of understanding and mitigating structural instabilities, specifically flutter, in aircraft.

The examination of combustion and propulsion mechanisms presented the ongoing advancements in propulsion technology. Systems designs emerged as the pivotal aspect of the sections we discussed as they control the aspects of aeroelasticity, aeroacoustics, and propulsion. We discussed the complexity of systems and how the complexity can be calculated and tracked for any given project. From there we delved into the rapidly evolving world of UAVs, discussing the history and emphasizing the applications in various situations from reconnaissance to package delivery.

Ultimately, the synthesis of these diverse topics not only highlighted the past and current state of aerospace engineering but also pointed towards future directions of the field. The interdisciplinary nature necessitates ongoing collaboration and innovation that come with pushing boundaries will continue to encourage advancements in the field for year to come

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